

Skills Summary

Arithmetic Sequences as Linear Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Recursive rule from a listed sequence (recursive-from-sequence)

A recursive formula always has two lines:

$$a_1 = \text{first term} \quad a_n = a_{n-1} + d \quad \text{for } n \geq 2$$

d is the **common difference**: $d = a_2 - a_1 = a_3 - a_2 = \dots$. Check at least two pairs — if the differences disagree, the sequence is not arithmetic. Reaching a_n from a_1 takes $n - 1$ steps, not n .

Example: Write the recursive formula for 6, 10, 14, 18, ... and find a_7 .

Step 1. The terms are listed, so test for a constant difference first — that is what licenses a recursive arithmetic rule.

Step 2. $10 - 6 = 4$ and $14 - 10 = 4$, so $a_1 = 6$, $a_n = a_{n-1} + 4$; then $a_7 = 6 + 6(4) = \mathbf{30}$.

Try it: Write the recursive formula for 9, 16, 23, ... and find a_5 .

check: 37

Skill 2 — Recursive rule from a situation (recursive-from-context)

Translating a story:

“starts at / already has” $\rightarrow a_1$ “each additional / every hour after” $\rightarrow d$

Carry the **units** into both lines, and say what n counts (hours, chairs, weeks). If the starting amount happens at step 1, then step n is $n - 1$ additions later.

Example: A seedling is 12 cm tall on day 1 and grows 5 cm each day after that. How tall is it on day 9?

Step 1. “Is 12 cm on day 1” names a_1 and “grows 5 cm each day” names d , so the situation is arithmetic and recursive.

Step 2. $a_1 = 12$, $a_n = a_{n-1} + 5$; day 9 is $9 - 1 = 8$ growth days, so $a_9 = 12 + 8(5) = \mathbf{52}$ cm.

Try it: A jar holds 30 marbles on day 1 and 7 marbles are added each day after that. How many marbles on day 6?

check: 59

Skill 3 — Reaching a later term (term-from-recursive)

Rule:

Given a_1 and $a_n = a_{n-1} + d$, the n th term is

$$a_n = a_1 + (n - 1)d.$$

Adding d term by term always works; the $(n - 1)$ shortcut is the same thing done all at once. A decreasing sequence simply has $d < 0$.

Example: For $a_1 = 3$ and $a_n = a_{n-1} + 8$, find a_{12} .

Step 1. Listing twelve terms is slow, so count the steps instead and use $a_1 + (n - 1)d$.

Step 2. From term 1 to term 12 is $12 - 1 = 11$ steps of 8: $a_{12} = 3 + 11(8) = \mathbf{91}$.

Try it: For $a_1 = 100$ and $a_n = a_{n-1} - 6$, find a_{10} .

check: 49

Skill 4 — Recursive $\rightarrow$ explicit linear rule (recursive-to-explicit)

Same sequence, two views:

$$\text{recursive } a_n = a_{n-1} + d \quad \Longleftrightarrow \quad \text{explicit } a_n = dn + (a_1 - d)$$

The common difference d is the **slope**, so an arithmetic sequence is a linear function of n . The intercept is $a_1 - d$, the value one step *before* the first term.

Example: Write $a_1 = 5$, $a_n = a_{n-1} + 7$ as an explicit rule and find a_{30} .

Step 1. The question asks for a far-out term, so convert the step rule into a linear function of n rather than iterating.

Step 2. $a_n = 5 + (n - 1)7 = 7n - 2$, so $a_{30} = 7(30) - 2 = \mathbf{208}$.

Try it: Write $a_1 = 11$, $a_n = a_{n-1} + 4$ explicitly and find a_{25} .

check: 101

(!) Watch out: The commonest error is multiplying by n instead of $n - 1$. The first term costs no steps: reaching term 5 takes 4 additions.